CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc (Biotechnology/Microbiology) Examination March 2018

Course code & Name: MI 812 Environmental Microbiology

Date: 29/03/2018 Day Thursday Time: 1:30 P M To1:50 P M Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **10**

1. The following is a quantitatively measurable change in living organisms at the cellular, biochemical, molecular or physiological levels upon exposure to pollutant
(i) bioindicator (ii) biomarker (iii) biomonitor (iv) biosensor

2. The following term describes the unavailability of organic pollutant for bioremediation
(i) sequesteration (ii) biodegradation (iii) cometabolism (iv) mineralization

3. The following term describes the chemical transformation of a substance caused by microbes or enzymes.
(i) bioaccumulation (ii) mineralization
(iii) biodegradation (iv) biotransformation

4. An increase in level of the following in microbial population is observed in the presence of metal ions
(i) superoxide dismutase (ii) LME (iii) monoxygenase (iv) carcinogen

5. A consortium producing the following enzyme(s) is more suitable for the biodegradation of effluent from dye industry

(i) azoreductase (ii) LME

(iii) nitrogenase and LME (iv) azoreductase and LME

6. A microorganism producing the following compound is more suitable for biodegradation of hydrocarbon, as it helps to emulsify HC

(i) siderophore (ii) biostimulant (iii) biosurfactant (iv) biopesticide

7. Introduction of air into the saturated zone below the water table for bioremediation of ground water is called

(i) bioventing (ii) biosparging (iii) bioaugmentation (iv) aeration

8. The growth rate (µn) of nitrifying bacteria must be higher than the growth rate (µh) of heterotrophs for the following

(i) nitrification (ii) denitrification (iii) ammonification (iv) none of the above

9. The following term indicates mean cell residence time in the activated sludge system

(i) SRT (ii) HRT (iii) F/M (iv) MLSS

10. The following process involves formation of biofilm for the degradation of pollutants present in industrial effluents

(i) Trickling filters (ii) activated sludge process

(iii) composting (iv) all of the above

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Course code & Name: MI 812 Environmental Microbiology

Date: 29/03/2018 Day Thursday Time: 1:50 P M To 3:30 P M Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed **10**

A Discuss briefly any one of the following **(04)**

i. Intracellular metal resistance mechanisms in microbes

ii. The aerobic and anaerobic mode of biodegradarion of aromatic hydrocarbons

B Attempt any two of the following: **(04)**

i. Write a short note on:" The fate of metals in environment"

ii. Explain the term co-metabolism for the biodegradation of environmental pollutant with example

iii. Discuss briefly the role of LME in biodegradation of organic pollutants

C Justify any one of the following **(02)**

i. F/M ratio should be maintained between 0.2 to 0.5 in activated sludge process

ii. Alkanes having 10 to 24 carbon atoms (C_{10}-C_{24}) are usually the easiest to be metabolized by microorganisms

SECTION - II

Q-III Answer the following questions as directed **15**

A Discuss briefly Any Two of the following **(06)**

i. Biosensors used in monitoring of BOD

ii. Sampling process in environmental monitoring

iii. Microbial community in activated sludge process

B Write a short note on any three of the following **(06)**

i. Heat shock proteins as molecular markers for monitoring environment

ii. Use of microorganisms as bioindicators

iii. Trickling filters for treatment of waste water

iv. *In situ* ground water biorestoration

C Attempt any one of the following **(03)**

i. Give a brief account of microbial community involved in composting of solid waste

ii. Give a brief account of microbial community in anaerobic digestion of sludge